Algebra, equations and inequalities

Lecture and worked examples

Brackets and distributivity

After a $+$ sign, removing the brackets keeps the signs of the terms. After a $-$ sign, every term inside changes sign. This rule applies to a sum or a difference of terms.

Brackets and distributivity

Remove the brackets and simplify: $A = x + (2 + 3x)$; $B = x - (2 + 3x)$; $C = -2x + (-4 + 5x)$; $D = -2x - (-4 + 5x)$.

Worked answer: Brackets and distributivity

$A = 4x + 2$; $B = -2x - 2$; $C = 3x - 4$; $D = -7x + 4$.

Brackets and distributivity

For all real numbers a, b, c, d:

$$b(c+d) = bc + bd, \qquad (a+b)(c+d) = ac + ad + bc + bd.$$

Brackets and distributivity

Expand and simplify:

$$A = -4(8x+7) + (-8x+5)(5x+9),$$
$$B = (6x+9)(-x+7) + (3x+5)(-9x+6),$$
$$C = -9(-x-7) - (-x+3)(-8x-8),$$
$$D = -5(-6x-7) + (x-7)(5x+8).$$

Worked answer: Brackets and distributivity

Expand each product, then collect terms of the same degree:

$$A = -32x - 28 - 40x^2 - 72x + 25x + 45 = -40x^2 - 79x + 17,$$
$$B = -6x^2 + 33x + 63 - 27x^2 - 27x + 30 = -33x^2 + 6x + 93,$$
$$C = 9x + 63 - (8x^2 - 16x - 24) = -8x^2 + 25x + 87,$$
$$D = 30x + 35 + 5x^2 - 27x - 56 = 5x^2 + 3x - 21.$$

Factorising with a common factor

$ka + kb = k(a + b)$ and $ka - kb = k(a - b)$. Factorising turns a sum into a product; expanding does the reverse.

Factorising with a common factor

Factorise:

$$A = 15x^2 - 21x,$$
$$B = 3(2 + 3x) - (5 + 2x)(2 + 3x),$$
$$C = (4x - 1)(x + 6) + (4x - 1),$$
$$D = (1 - 6x)^2 - (1 - 6x)(2 + 5x).$$

Worked answer: Factorising with a common factor

$A = 3x(5x - 7)$.

$B = (2 + 3x)[3 - (5 + 2x)] = (2 + 3x)(-2 - 2x) = -2(2 + 3x)(1 + x)$.

$C = (4x - 1)[(x + 6) + 1] = (4x - 1)(x + 7)$.

$D = (1 - 6x)[(1 - 6x) - (2 + 5x)] = (1 - 6x)(-1 - 11x) = (6x - 1)(11x + 1)$. Check the factors by expanding again.

Factorising with a common factor

Choose a useful form of $f(x) = (x-2)(x+3)$ to calculate $f(2)$, $f(0)$ and find its zeros.

Worked answer: Factorising with a common factor

The factorised form immediately gives $f(2) = 0$ and the zeros $2, -3$. At 0, $f(0) = (-2) \times 3 = -6$; the expanded form $x^2 + x - 6$ also gives this result. The choice depends on the question.

Special identities: expanding

$$(a+b)^2 = a^2 + 2ab + b^2, \qquad (a-b)^2 = a^2 - 2ab + b^2,$$
$$(a+b)(a-b) = a^2 - b^2.$$

Special identities: expanding

Expand $(a+b)(a+b)$: $a^2+ab+ba+b^2 = a^2+2ab+b^2$. Replacing b by $-b$ gives the second identity. Finally $(a+b)(a-b) = a^2-ab+ab-b^2 = a^2-b^2$.

Special identities: expanding

Expand $(x+3)^2$, $(3x-4)^2$, $(x-3)(x+3)$, $(2x+3)(2x-3)$, $(4-3x)(3x+4)$.

Worked answer: Special identities: expanding

$x^2 + 6x + 9$; $9x^2 - 24x + 16$; $x^2 - 9$; $4x^2 - 9$; $16 - 9x^2$. In the last product, reorder $3x + 4$ as $4 + 3x$.

Special identities: expanding

Expand and simplify $A = (2x-3)^2 + (x+5)(3-x)$ and $B = 2(x+3) + (2x+3)(2x-3)$.

Worked answer: Special identities: expanding

$A = 4x^2 - 12x + 9 - x^2 - 2x + 15 = 3x^2 - 14x + 24$. $B = 2x + 6 + 4x^2 - 9 = 4x^2 + 2x - 3$.

Special identities: factorising

Spot the two squares; for three-term forms, check the double product $2ab$. For a difference of two squares, use $a^2 - b^2 = (a - b)(a + b)$.

Special identities: factorising

Factorise:

$$A = x^2 - 2x + 1, \quad B = 25 + 16x^2 - 40x, \quad C = 4x^2 + 12x + 9,$$

$$D = 1 - 49x^2, \quad E = 9x^2 - 4, \quad F = (2x + 3)^2 - 64.$$

Worked answer: Special identities: factorising

$A = (x-1)^2$. $B = (4x-5)^2$. $C = (2x+3)^2$.

$D = (1-7x)(1+7x)$. $E = (3x-2)(3x+2)$.

$F = [(2x+3)-8][(2x+3)+8] = (2x-5)(2x+11)$. For the first three, check the middle term; for F, treat $2x+3$ as a single block.

Special identities: factorising

Are the equalities $(a+b)^2 = a^2 + b^2$ and $a^2 + b^2 = (a-b)(a+b)$ true for all real numbers?

Worked answer: Special identities: factorising

No: for $a = b = 1$, the first would give $4 = 2$ and the second $2 = 0$. The missing term in the first is $2ab$; the product in the second equals $a^2 - b^2$.

Integer powers

For $n \geq 1$, a^n is the product of n factors equal to a. For $a \neq 0$, $a^0 = 1$ and $a^{-n} = 1/a^n$. In this course, 0^0 is not defined. For $a, b \neq 0$ and integers m, n:

$$a^m a^n = a^{m+n}, \quad \frac{a^m}{a^n} = a^{m-n}, \quad (a^m)^n = a^{mn},$$

$$(ab)^n = a^n b^n, \qquad \left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}.$$

For strictly positive exponents, some bases may be zero if the expressions remain defined.

Integer powers

Simplify $A = 2^4 \times 2^3$, $B = (-5)^{12}/(-5)^3$, $C = (7^3)^6 \times 7^{-5}$, $D = 7^{24} \times 3^2 \times 2^3$. Also calculate $(-3)^5$, 5^1 and 99999^0.

Worked answer: Integer powers

$A = 2^7 = 128$. $B = (-5)^9 = -1953125$. $C = 7^{18-5} = 7^{13}$. $D = 72 \times 7^{24}$: exponents of different bases cannot be added. The last three values are -243, 5, 1.

Integer powers

The definition as a repeated product does not hold for a negative exponent. $-3^2 = -9$, whereas $(-3)^2 = 9$.

Integer powers

Solve $2^n > 100$ mentally for the smallest natural number n. Write a program that finds it.

Worked answer: Integer powers

$2^6 = 64 \leq 100 < 128 = 2^7$, so $n = 7$. Initialise $n = 0$, $p = 1$; while $p \leq 100$, replace p by $2p$ and n by $n + 1$. The program is given at the end of the sequence.

For nonzero denominators:

$$\frac{a}{c} \pm \frac{b}{c} = \frac{a \pm b}{c}, \quad \frac{a}{c} \pm \frac{b}{d} = \frac{ad \pm bc}{cd}, \quad \frac{a}{c} \times \frac{b}{d} = \frac{ab}{cd}.$$

A fraction c/d has an inverse only if $c \neq 0$ and $d \neq 0$. Then

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc} \quad (b, c, d \neq 0).$$

Calculate and reduce to lowest terms:

$$A = \left(\frac{2}{3} + \frac{1}{6}\right) \times \frac{4}{5}, \quad B = \frac{1}{2} + \frac{3}{4} \times \frac{5}{6}, \quad C = \frac{4}{5} \div \frac{2}{3} \times \frac{7}{10},$$

$$D = \frac{\left(\frac{2}{5} + \frac{1}{3}\right) \times \frac{3}{4}}{\frac{1}{2} - \frac{1}{6}}.$$

Worked answer: Fractions and fractional expressions

$A = \frac{5}{6} \times \frac{4}{5} = \frac{2}{3}$. $B = \frac{1}{2} + \frac{5}{8} = \frac{9}{8}$. $C = \frac{4}{5} \times \frac{3}{2} \times \frac{7}{10} = \frac{21}{25}$. $D = \frac{11}{15} \times \frac{3}{4} \div \frac{1}{3} = \frac{11}{20} \times 3 = \frac{33}{20}$.
In C, multiplication and division are carried out from left to right.

Simplify $\dfrac{x^2-1}{x-1}$, stating the domain. Can we then take $x=1$?

Worked answer: Fractions and fractional expressions

For $x \neq 1$, $\dfrac{(x-1)(x+1)}{x-1} = x+1$. The original expression remains undefined at 1; the simplification does not lift the restriction.

Square roots and validity conditions

For $a \geq 0$, $\sqrt{a}$ is the unique positive or zero number whose square is a. Thus $(\sqrt{a})^2 = a$. For every $x \in \mathbb{R}$, $\sqrt{x^2} = |x|$. For $a, b \geq 0$, $\sqrt{ab} = \sqrt{a}\sqrt{b}$. If $b > 0$, $\sqrt{a/b} = \sqrt{a}/\sqrt{b}$.

Square roots and validity conditions

The number $\sqrt{a}\sqrt{b}$ is positive or zero and its square is ab; by uniqueness of the positive square root, it equals $\sqrt{ab}$. The same argument applied to the quotient establishes the second formula, under $b > 0$. Similarly $|x| \geq 0$ and $|x|^2 = x^2$, so $\sqrt{x^2} = |x|$.

Square roots and validity conditions

Simplify $\sqrt{18}$; $\sqrt{2}\sqrt{8}$; $\sqrt{7/25}$; $\sqrt{27/75}$; $\sqrt{(-5)^2}$.

Worked answer: Square roots and validity conditions

$3\sqrt{2}$; 4 ; $\sqrt{7}/5$; $\sqrt{9/25} = 3/5$; 5.

Square roots and validity conditions

There is no general rule $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$. With $a = 4, b = 9$, we would wrongly have $\sqrt{13} = 5$. Saying these expressions are always different would also be wrong, for example if $a = 0$.

Square roots and validity conditions

Simplify $2\sqrt{18} - \sqrt{50} + \sqrt{8}$. Bound $\sqrt{20}$ between two consecutive integers.

Worked answer: Square roots and validity conditions

$6\sqrt{2} - 5\sqrt{2} + 2\sqrt{2} = 3\sqrt{2}$. Since $16 < 20 < 25$, $4 < \sqrt{20} < 5$.

First-degree equations

To solve an equation is to find all the values of the unknown that satisfy the equality. Two equations are equivalent if they have the same solutions. Adding the same expression or multiplying by the same nonzero number preserves equivalence.

First-degree equations

Expand if necessary, collect the terms in x, then isolate x. If its coefficient is zero, examine the remaining equality: it may be always true or impossible. Check in the original equation.

First-degree equations

Solve:

$a)\ 5x-3=2x+9,$ $b)\ 3(2x-1)=2(x+4)+7,$

$c)\ -2(x+3)=4x-(x-1),$ $d)\ \dfrac{x+2}{3}=x-1,$

$e)\ \dfrac{-5x+5}{8}=-\dfrac{3}{4},$ $f)\ \dfrac{-3x-1}{9}=-\dfrac{7}{4}.$

Worked answer: First-degree equations

a) $3x = 12$, so $x = 4$. b) $6x - 3 = 2x + 15$, so $x = 9/2$. c) $-2x - 6 = 3x + 1$, so $x = -7/5$. d) $x + 2 = 3x - 3$, so $x = 5/2$. e) $-5x + 5 = -6$, so $x = 11/5$. f) $4(-3x - 1) = -63$, so $-12x = -59$ and $x = 59/12$.

Solve $2(x+1) = 2x+2$, then $2(x+1) = 2x+3$.

Worked answer: First-degree equations

The first reduces to $2 = 2$: $S = \mathbb{R}$. The second gives $2 = 3$: $S = \varnothing$. We do not divide by a zero coefficient.

Zero-product property and equations with a quotient

$AB = 0 \iff A = 0$ or $B = 0$. For $B \neq 0$, $A/B = 0 \iff A = 0$. To solve $A(x)/B(x) = k$, first require $B(x) \neq 0$, then solve $A(x) = kB(x)$ and discard the forbidden values.

Zero-product property and equations with a quotient

Solve $(4x+6)(3-7x)=0$; $3x(-7x+5)=0$; $\dfrac{2x+1}{3x-5}=0$.

Worked answer: Zero-product property and equations with a quotient

Solutions: $\{-3/2, 3/7\}$; $\{0, 5/7\}$; $\{-1/2\}$, with forbidden value $5/3$ in the last case.

Zero-product property and equations with a quotient

Factorise then solve $(1-6x)^2-(1-6x)(2+5x)=0$ and $(4x-1)(x+6)+(4x-1)=0$.

Worked answer: Zero-product property and equations with a quotient

First: $(6x-1)(11x+1)=0$, so $S=\{1/6,-1/11\}$. Second: $(4x-1)(x+7)=0$, so $S=\{1/4,-7\}$.

Zero-product property and equations with a quotient

Solve $\dfrac{x+1}{x-2} = 3$, then $\dfrac{x^2-4}{x-2} = 4$.

Worked answer: Zero-product property and equations with a quotient

In both cases, $x \neq 2$. First: $x + 1 = 3x - 6$, so $x = 7/2$, allowed. Second: for $x \neq 2$, the quotient equals $x + 2$; the equation requires $x = 2$, forbidden. So no solution.

Equations with a square; isolating a variable

In $\mathbb{R}$, $x^2 = a$ has two solutions $\pm\sqrt{a}$ if $a > 0$, one solution 0 if $a = 0$, and none if $a < 0$.

Equations with a square; isolating a variable

Solve $x^2 = 0$; $x^2 = -8$; $x^2 = 7$; $25x^2 = 81$; $(x+2)^2 = 9$.

Worked answer: Equations with a square; isolating a variable

In order: $\{0\}$; $\varnothing$; $\{-\sqrt{7}, \sqrt{7}\}$; $\{-9/5, 9/5\}$; $\{-5, 1\}$. For the last one, $x+2 = -3$ or $x+2 = 3$.

Equations with a square; isolating a variable

Describe the operations performed on both sides; state the conditions that allow dividing or taking a square root.

Equations with a square; isolating a variable

Isolate h in $V = \pi r^2 h$ with $r > 0$; isolate v in $E = \frac{1}{2}mv^2$ with $m > 0$, $E \geq 0$ and $v \geq 0$; isolate x in $y = ax + b$ with $a \neq 0$.

Worked answer: Equations with a square; isolating a variable

$h = V/(\pi r^2)$. Then $v^2 = 2E/m$; since $v \geq 0$, $v = \sqrt{2E/m}$. Finally $x = (y - b)/a$. Without the condition $v \geq 0$, two signs would be possible for $E > 0$.

Inequalities and comparison

Inequalities of the same direction can be added: $a \leq b$, $c \leq d \Rightarrow a + c \leq b + d$. Adding the same real number preserves the direction. Multiplying or dividing by a strictly positive real number preserves the direction; by a strictly negative one, it reverses it. Comparing A and B amounts to studying the sign of $A - B$. If $A, B > 0$, one can also compare A/B to 1.

1. If $x \leq -4$, what can be said about $3x - 4$? 2. If $x > -6$, what can be said about $2x - 3$? 3. If $7 \leq x \leq 16$ and $3 \leq y \leq 12$, bound $8x + 7y$. 4. If $-10 \leq x \leq -6$, bound $2x - 6$.

Worked answer: Inequalities and comparison

1. $3x - 4 \leq -16$. 2. $2x - 3 > -15$. 3. $56 \leq 8x \leq 128$ and $21 \leq 7y \leq 84$, so $77 \leq 8x + 7y \leq 212$. The bounds are attainable. 4. $-26 \leq 2x - 6 \leq -18$.

Solve $3x + 1 \geq x - 7$; $2x - 11 \leq 4x - 5$; $6x + 19 \leq 2x - 3$.

Worked answer: Inequalities and comparison

1. $2x \geq -8$, that is $[-4, +\infty)$. 2. $-6 \leq 2x$, that is $[-3, +\infty)$. 3. $4x \leq -22$, that is $(-\infty, -11/2]$.

Sign of an affine expression and of a product

For $a \neq 0$, $ax + b$ vanishes at $x_0 = -b/a$; its sign to the right is that of a, to the left the opposite. If $a = 0$, the sign is that of the constant b. For a product, multiply the signs of the factors.

Sign of an affine expression and of a product

Draw the sign tables of $3 - x$, $2x - 6$, then $P(x) = (2x - 6)(x + 1)$.

Worked answer: Sign of an affine expression and of a product

Both affine expressions vanish at 3: $3 - x$ is positive on the left, negative on the right; $2x - 6$ has the opposite signs.

x	$-\infty$		-1		3		$+\infty$
$2x-6$		$-$	$-$	$-$	0	$+$	
$x+1$		$-$	0	$+$	$+$	$+$	
$P(x)$		$+$	0	$-$	0	$+$	

Sign of an affine expression and of a product

Solve $(7 + 3x)(2x + 6) > 0$.

Worked answer: Sign of an affine expression and of a product

The zeros are $-7/3$ and -3, to be ordered $-3 < -7/3$. The product is positive outside these roots and negative between them. The inequality is strict: $S = (-\infty, -3) \cup (-7/3, +\infty)$.

Sign of a quotient and algorithmic synthesis

Find the forbidden values, then the zeros of the numerator. Order all the critical values and apply the sign rule. On the quotient row, a double bar indicates a forbidden value.

Sign of a quotient and algorithmic synthesis

Draw the sign table of $Q(x) = \dfrac{-4x+6}{x-3}$, then solve $\dfrac{2x-1}{-4x+3} \leq 0$.

Worked answer: Sign of a quotient and algorithmic synthesis

For Q, zero $3/2$, forbidden 3. Signs: negative on $(-\infty, 3/2)$, zero at $3/2$, positive on $(3/2, 3)$, undefined at 3, negative on $(3, +\infty)$. For the second expression, zero $1/2$, forbidden $3/4$. The signs are $-, +, -$ on the three intervals. Thus $S = (-\infty, 1/2] \cup (3/4, +\infty)$.

Python

```
def premiere_puissance(seuil):
    # seuil reel >= 1
    n, p = 0, 1
    while p <= seuil:
        n = n+1
        p = 2*p
    return n, p
```

Sign of a quotient and algorithmic synthesis

What does `premiere_puissance(100)` return? What relation between `p` and `n` stays true at each step? How can the sign of a number then be tested in Python?

Worked answer: Sign of a quotient and algorithmic synthesis

The result is $(7, 128)$. The invariant is $p = 2^n$, true at initialisation and then preserved. For the sign, first test `x > 0`, then `x < 0`, otherwise $x = 0$. Do not replace the loop's non-strict inequality by a strict one without changing the problem posed.